6.3.2 Intercepts of a Line

Definition 6.6

The **intercept** of a line is the point at which the line crosses either the x-axis or the y-axis.

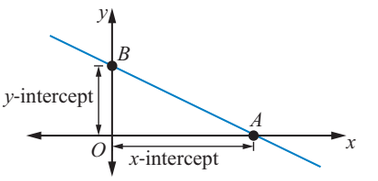


Figure 6.16

The x-intercept is a point where the y value is zero, and the y-intercept is a point where the x-value is zero.

Therefore the intercepts of a line are the points where the line intersects, or crosses, the horizontal and vertical axes.

Therefore it is clear that

(i) the equation of the y-axis is $x = 0$.

(ii) the equation of the x-axis is $y = 0$.

In the figure *OA* is the x-intercept and *OB* is the y-intercept.

Different types of *x* and *y* Intercepts:

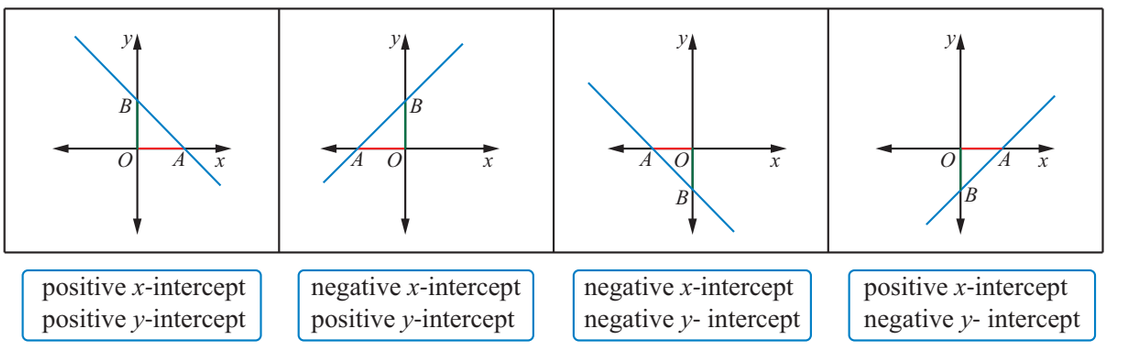


Figure 6.17

We have learnt the definition and detailed information about the points, slope and intercepts. Using these information, let us recall the different forms of an equation of a straight line.

6.3.3 Different Forms of an equation of a straight line

Two conditions are sufficient to determine uniquely the equation of a straight line. Using the combination of any two information from slope, intercepts and points, we can now form different types of straight lines such as

(i) Slope and intercept form
(ii) Point and Slope form
(iii) The two Point form
(iv) Intercepts form and two more special types are
(v) Normal form
(vi) Parametric form